HOT DATA FOR REAL-TIME AND RECENT DATA ACCESS

Hot data includes recent operational measurements, current configuration data, and frequently accessed reference information. This data requires high-performance storage with low-latency access, typically using premium storage tiers and memory caching.

WARM DATA FOR HISTORICAL ANALYSIS AND REPORTING

Data from weeks to months old serves different analytical purposes with relaxed performance requirements. Monthly reports, trend analysis, and model training workloads can tolerate query response times measured in seconds rather than milliseconds. This shift in performance expectations enables cost optimizations through compression, aggregation, and migration to slower but cheaper storage tiers.

COLD DATA FOR ARCHIVAL ACCESS

Historical data beyond immediate operational relevance can be moved to archival storage where access times measured in minutes or hours become acceptable. This cold data primarily serves compliance auditing, forensic analysis, and occasional deep historical studies that justify longer retrieval delays.

Cloud archival services like AWS Glacier or Azure Archive Tier provide cost-effective cold storage with retrieval times ranging from minutes to hours depending on the storage class selected. The dramatic cost reduction—often 10:1 or better compared to hot storage—justifies the performance trade-off for infrequently accessed data. Figure 4.9 shows an example of lifecycle tiering and the effect on storage cost for data stored in Amazon S3.

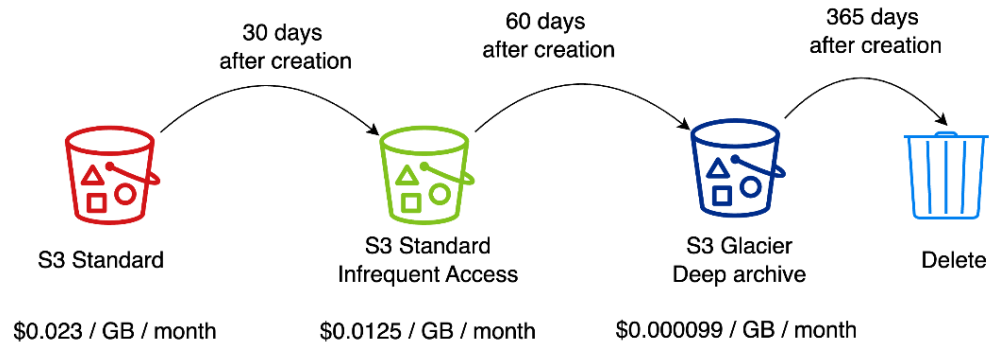


Figure 4.9 An example lifecycle policy for data stored in Amazon S3 that transitions data to cheaper storage and eventual deletion as it ages.

4.6 Data governance and compliance

Digital twins aggregate vast amounts of operational and business data from across organizational boundaries, creating both new analytical opportunities but also significant